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NPS/002/008 – Specification for Cold Pour Resins and Hot and Cold Pour Compounds

1. Purpose

The purpose of this document is to specify cold pour resin for use on cable joints up to 33kV, including PVC, waveform and PILC mains cables (including transition joints); concentric, split concentric and PILC service cables (including transition joints). Also hot pour compound for use within certain cable joints and terminations up to 66kV.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
DSS/005/013	1	Specification for resins and compounds
MS-U-21		Resin

2. Scope

This document applies to the specification required for cold pour resin and hot poured compounds used in YEDL and NEDL.

This document includes the required information to procure the above items.

3. Technical Requirements

3.1 Resin

The resin and compounds shall comply in all respects with the specification detailed below unless otherwise agreed in writing by a representative of NEDL/YEDL.

The method of supplying resin and hardener is by the use of a system which eliminates problems of hardener container disposal, and on completion of mixing any containers shall be considered non-hazardous and suitable for disposal as general waste.

The filling media shall be two-part cold pouring resin compound identical to that which Type Approval has been obtained in accordance with Engineering Recommendation C81 "Type Approval Tests for Joints for 600/1000 volt Cable Systems" or equivalent.

Resin and hardener shall be adequately protected and sealed against ingress of moisture.

The "setting time" (i.e. the time required for curing to reach a stage where normal back-filling may commence without damage to the joint) shall not exceed 60 minutes at 0°C and shall not be less than 20 minutes at 30°C.

Graphical values of resin mixing and setting times and of resin temperature variation with time at ambient temperatures between -5°C and 30°C shall be supplied with the tender.

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3.2 Re-Enterable Resin

Re-enterable resin shall comply with the general requirements of two part cold applied resin as stated above. It will however set in such a way that removal of the resin will be achievable with an insulated knife from the cable to allow any joints to be undone and remade.

3.3 Compounds

The compound as used for up to 20kV shall be identical to that which Type Approval has been obtained in accordance with BS 1858 2002 or equivalent.

The compound used for 33 to 66kV shall be identical to that which Type Approval has been obtained in accordance with BS 2000 institute of Petroleum Standards or equivalent.

Copies of all witnessed test reports on which Type Approval was given shall be included with the tender document.

The classification of compounds up to 20kV will be that as defined in table 1 of BS 1858 2002. The compound used for up to 11kV, shall be class 11 of BS 1858 2002. Compounds used for up to 20kV, shall be class 5 of BS 1858 2002.

The compounds shall have the general requirements of clause 4 of BS 1858 2002 and have characteristics as defined in clauses five through to 16 or equivalent.

The classification of compounds rated between 33 and 66kV shall be defined and tested in accordance with the Institute of Petroleum standard BS 2000.

3.4 Classification of use

Type One	For use in cable joints and terminations up to 11kV
Type Two	For use in cable joints and terminations up to 20kV
Type Three	For use in cable joints and terminations between 33 and 66kV.
Type Four	Two part Cold pour Resin or equivalent. For use in cable joints up to 33kV
Type Five	Generally complying with the requirements of a Two part Cold pour Resin or equivalent. For use in cable joints and terminations up to 33kV but have the ability to be removed from the cable joint or termination

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4. References

4.1 External Documentation

The Resin and Compound shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

Reference	Title
BS 1858 2002	Electric cables. Accessories. Bitumen-based filling compounds
Engineering Recommendation C81	Type Approval Tests for Joints for 600/1000 volt Cable Systems
BS 2000	Institute of Petroleum Standards

4.2 Internal documentation

Reference	Title
NIL	

5 Definitions

Term	Definition
NIL	

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6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Underground Technical Support	R McMahon	29/11/05

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Procurement Manager	G Bolden	29/11/05

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D M Gazda	Underground Standards and Assessments Manager	D M Gazda	29/11/05

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system.

		Sign	Date
S Johnson	DBD Administrator	S Johnson	23/12/05

6.5 Approval

Approval is given for the content of this document.

		Sign	Date
C. Holdsworth	Standards and Assessment Manager	C. Holdsworth	30/11/05

6.6 Authorisation

Authorisation is granted for publication of this document.

		Sign	Date
C. Morgan	Head of Procurement	C. Morgan	01/12/05

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Appendix 1 – Current Requirements & Catalogue Numbers

Description	NEDL/YEDL Catalogue Number
Compound (A57)	160996
Compound (G38a)	160939
2 Litre Resin Pack (2 x 2 Litre)	085464
4 Litre Resin Pack (2 x 4 Litre)	085464
Re-enterable Resin Pack (1.85 Litre)	085316

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Appendix 2 – Addendum to Supplier Requirements

Copies of all witnessed test reports on which Type Approval was given shall be included with the tender document.

Suitable protective gloves or mittens shall be supplied with each resin pack.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

All packages shall include written details on how to mix and handle the Resin/Compound and list any special precautions required for use with this product.

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Appendix 3 – Logistical Requirements

To enable NEDL/YEDL to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

Components shall be supplied complete in a package designed to withstand general handling, transport and storage without damage to the components and adequately sealed against moisture. This package must contain sufficient resin, hardener and protective gloves or mittens required. Alternative pack sizes will be considered.

The package should be clearly labelled with the YEDL/NEDL reference numbers, the shelf life expiry date and the manufacturer's batch number. The shelf life, expiry date should be that of the item within the package, which has the minimum life, i.e. resin or hardener etc.

This package must contain sufficient resin, hardener and protective gloves or mittens required. Alternative pack sizes will be considered.

The package should be clearly labelled with the YEDL/NEDL reference numbers, the shelf life expiry date and the manufacturer's batch number. The shelf life, expiry date should be that of the item within the package, which has the minimum life, i.e. resin or hardener etc.

The method for the supply and mixing of the resin is by the use of a system, which eliminates problems of hardener container disposal, and on completion of mixing shall be considered non-hazardous and suitable for disposal as general waste.

Each component shall be suitably labelled to show: -

- The type and reference number of the resin/compound
- Volume and/or weight
- Shelf life expiry date where applicable
- Batch number
- Any safety warning or advice required by Statute or considered necessary by the manufacturer or YEDL/NEDL

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.